

RW3 Module Report - model_forecast_knn

INTERPRETATION GUIDE

- ADF $p > 0.05$ and KPSS $p < 0.05$ -> likely $I(1)$ -like; prefer differences/cointegration models.
- ADF $p < 0.05$ and KPSS $p > 0.05$ -> stationary; level models are more admissible.
- Ljung-Box $p < 0.05$ -> autocorrelation; add lag structure.
- BP/White $p < 0.05$ -> heteroskedasticity; use robust or HAC standard errors.
- JB $p < 0.05$ -> non-normality; use robust inference and avoid strict normality assumptions.
- Stability flag=1 -> potential breaks/drift; use rolling or split-sample robustness.
- For retail transmission, compare no-promo vs promo-controlled estimates.

RUN NOTES

module=model_forecast_knn

xlsx_files=1

png_files=3

Interpretation: read test diagnostics first, then coefficients/effects with robustness context.

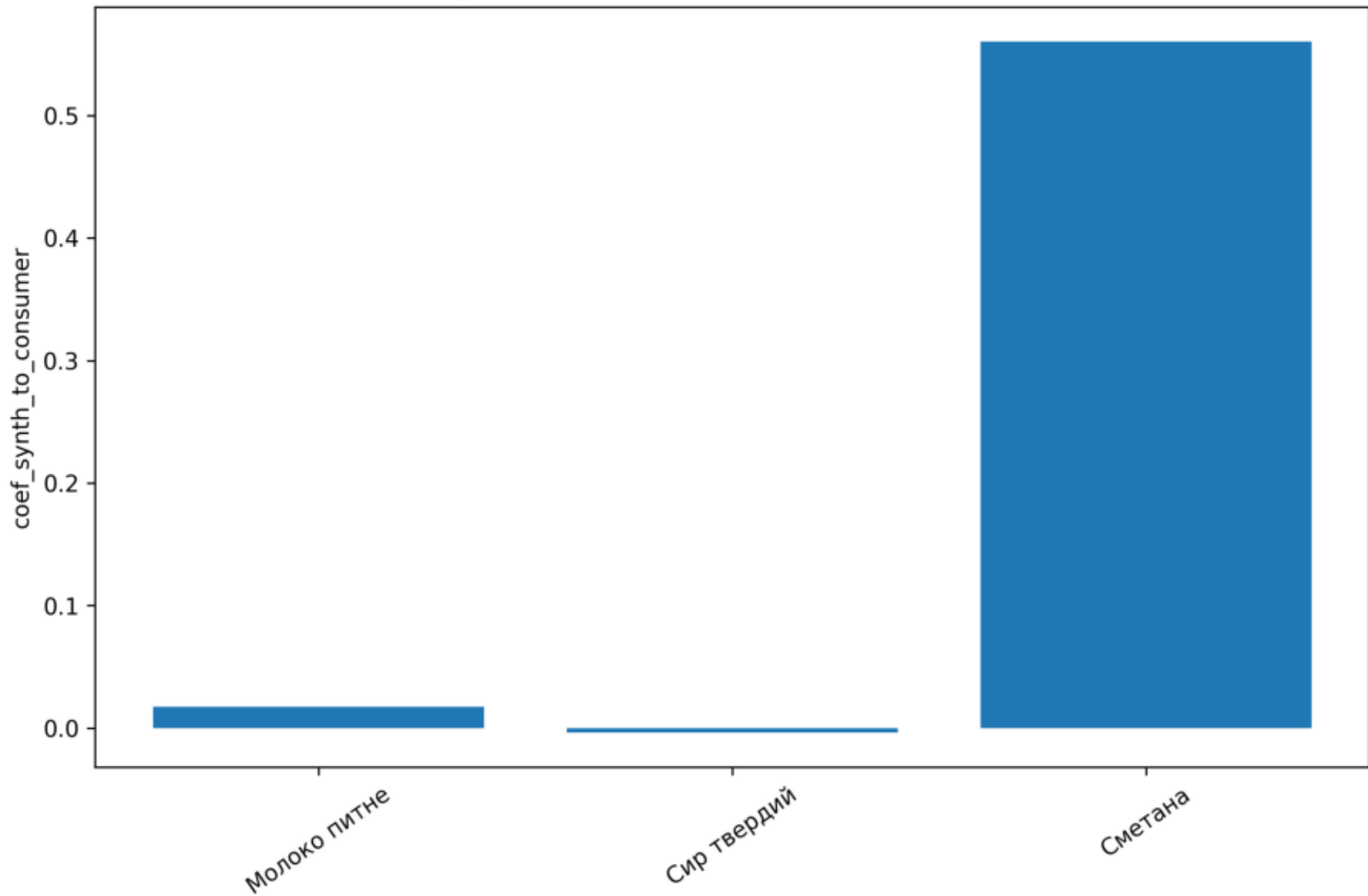
Table: model_forecast_knn_output_Forec

date	product	target	actual_dlog	pred_dlog	segment
2025-11-30	Вершки	ProducerUA	-0.03093304111750506	-0.03774108873871674	holdout
2025-12-01	Вершки	ProducerUA	-0.01224326412504606	-0.02856545768700842	holdout
2025-12-02	Вершки	ProducerUA	8.740190458889074e-05	-0.01131223770835968	holdout
2025-12-03	Вершки	ProducerUA	0.000225369664060171	7.065324739551072e-05	holdout
2025-12-04	Вершки	ProducerUA	0.0003080763376743789	0.000198016356387113	holdout
2025-12-05	Вершки	ProducerUA	0.0003355572578493948	0.0002743659262157945	holdout
2025-12-06	Вершки	ProducerUA	0.0003078781437038103	0.0002997345735349433	holdout
2025-12-07	Вершки	ProducerUA	0.0002251046165397952	0.0002741829660780656	holdout
2025-12-08	Вершки	ProducerUA	8.727183181589382e-05	0.0001977716812807335	holdout
2025-12-09	Вершки	ProducerUA	-0.0002736795051725416	7.053317243866912e-05	holdout
2025-12-10	Вершки	ProducerUA	-0.0007633284874453139	-0.0002626742714576152	holdout
2025-12-11	Вершки	ProducerUA	-0.001166757175254496	-0.0007146872412689858	holdout
2025-12-12	Вершки	ProducerUA	-0.001484126713985567	-0.001087107088722374	holdout
2025-12-13	Вершки	ProducerUA	-0.001715286797824511	-0.001380082573045102	holdout
2025-12-14	Вершки	ProducerUA	-0.001859848815486398	-0.001593474941416612	holdout
2025-12-15	Вершки	ProducerUA	-0.001917259080641998	-0.001726925452478579	holdout
2025-12-16	Вершки	ProducerUA	-0.002775386477930297	-0.001779922978032884	holdout
2025-12-17	Вершки	ProducerUA	-0.00408316942382303	-0.002572091914082473	holdout
2025-12-18	Вершки	ProducerUA	-0.004773150179377073	-0.003779354411592024	holdout
2025-12-19	Вершки	ProducerUA	-0.004836039753266519	-0.004416301003379148	holdout
2025-12-20	Вершки	ProducerUA	-0.004260078144329604	-0.00447435668018663	holdout
2025-12-21	Вершки	ProducerUA	-0.00303503148019324	-0.0039426653527698	holdout
2025-12-22	Вершки	ProducerUA	-0.001156414675201667	-0.002811779757453038	holdout
2025-12-23	Вершки	ProducerUA	0.0001420316153999401	-0.001077559546843589	holdout
2025-12-24	Вершки	ProducerUA	0.0005005095647452329	0.0001210839410508336	holdout
2025-12-25	Вершки	ProducerUA	0.0009701468606420072	0.0004520081099183773	holdout
2025-12-26	Вершки	ProducerUA	0.001549942937089988	0.0008855475566198611	holdout
2025-12-27	Вершки	ProducerUA	0.002238321257383546	0.001420778621910293	holdout
2025-12-28	Вершки	ProducerUA	0.003033020901482164	0.002056245946820032	holdout
2025-12-29	Вершки	ProducerUA	0.003931000006233365	0.00278986239297236	holdout
2025-12-30	Вершки	ProducerUA	nan	0.003618819904704773	future_30d
2025-12-31	Вершки	ProducerUA	nan	0.00333063498057203	future_30d
2026-01-01	Вершки	ProducerUA	nan	0.003064600886500251	future_30d
2026-01-02	Вершки	ProducerUA	nan	0.002819015037686062	future_30d
2026-01-03	Вершки	ProducerUA	nan	0.002592305715528606	future_30d
2026-01-04	Вершки	ProducerUA	nan	0.002383022008827819	future_30d
2026-01-05	Вершки	ProducerUA	nan	0.002189824528134893	future_30d
2026-01-06	Вершки	ProducerUA	nan	0.002011476833827962	future_30d
2026-01-07	Вершки	ProducerUA	nan	0.001846837523053744	future_30d
2026-01-08	Вершки	ProducerUA	nan	0.001694852924892577	future_30d

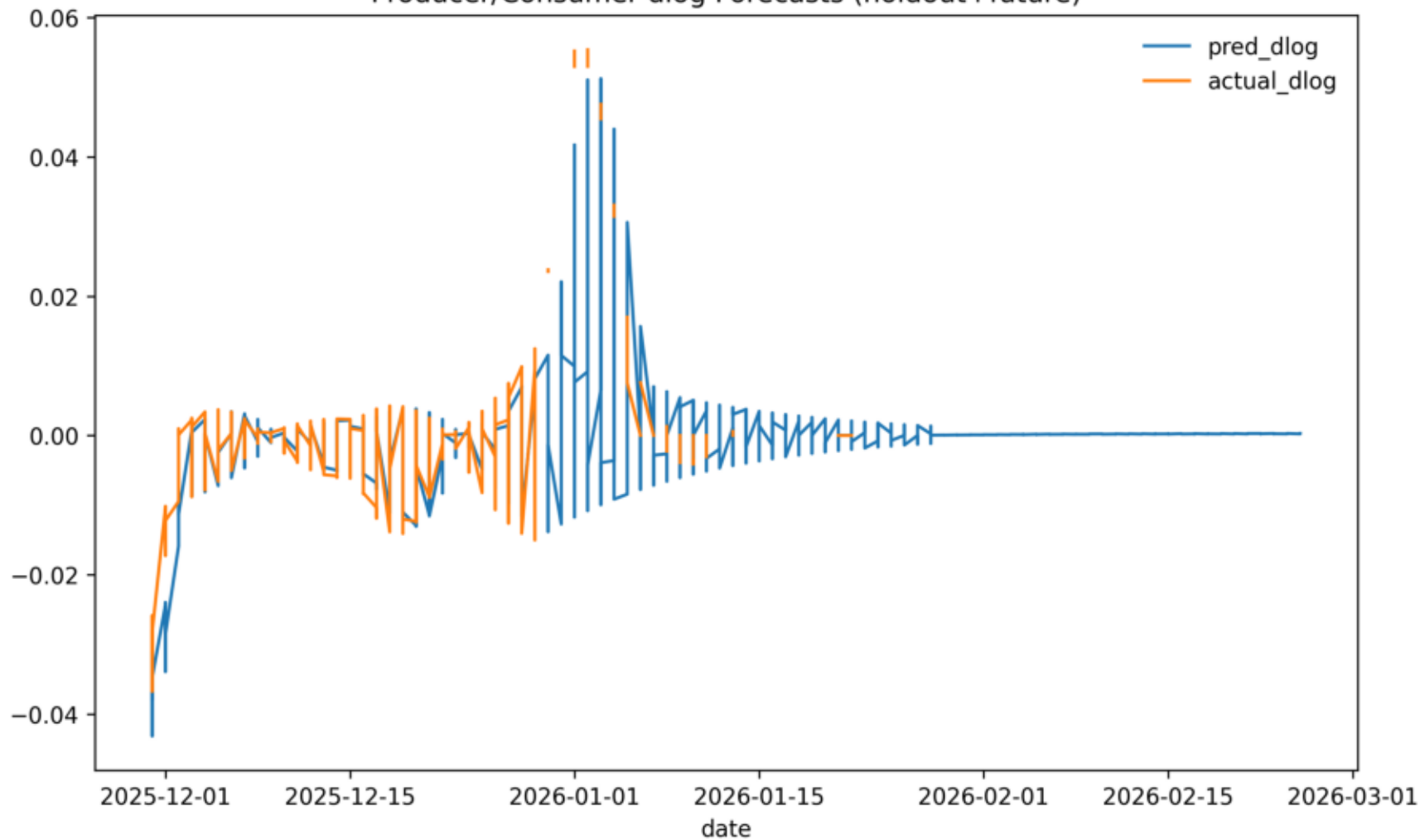
Table: model_forecast_knn_output_Synth

product	n_obs	coef_synth to consumer	p_synth to consumer	coef_producer to consumer	p_producer to consumer	r2
Молоко питне	50	0.01753647005461285	0.1790711340439692	-0.1539373703382407	0.8758089782929801	0.5682257019300552
Сир твердий	46	-0.003448418725580535	0.393544581866999	-0.1304863006531816	0.3633535989144401	0.8530481539621354
Сметана	49	0.56040945620913	0.08992558573282051	-5.078263514758259	0.1164214278893085	0.4607852937005169

Synthetic Retail -> ConsumerUA Influence



Producer/Consumer dlog Forecasts (holdout+future)



Synthetic Retail Price (Top product-brand)

